

Student Name: (Please Print)	
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 $\mathbb{Z}_{26}$ encoding:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

m -bit LFSR: $k_{m+i} = \sum_{j=0}^{m-1} c_j k_{j+i} \bmod 2$ where $i \geq 1$, c_j are constants, and $(k_1, k_2, \dots, k_m)$ are the initial bits.

RC4 keystream:

```
#initialization
for i in range(n):
    S[i]=i
    T[i]=key[i%len(key)]

#initial permutation of S
j=0
for i in range(n):
    j=(j+S[i]+T[i])%n
    S[i],S[j]=S[j],S[i] #swap
```

```
#keystream generation
i=0
j=0
while True:
    i=(i+1)%n
    j=(j+S[i])%n
    S[i],S[j]=S[j],S[i] #swap
    t=(S[i]+S[j])%n
    print(S[t])
```

Feistel Cipher:

